TEACHING STATEMENT

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My first experience teaching in a formal setting was during my early undergraduate years. I served as an undergraduate teaching assistant in the Department of Information Studies at the University of Ghana. I supported students in programming, information systems, and database concepts, and I led training sessions in Visual Basic .NET. This was a challenging experience for me. I had to keep students with very different levels of background interested and engaged. After moving to the United States for graduate study, I took up teaching enthusiastically as a graduate teaching assistant at East Tennessee State University. There I led sessions in introductory statistics, linear algebra, pre-calculus, and probability and statistics. I also set and graded work for these courses.

During my graduate years, I gave a number of talks at conferences and continued to teach. At the University of Cincinnati, I spent two years as a graduate teaching assistant in the Department of Mathematical Sciences, teaching quantitative reasoning, introductory statistics, and calculus problem sessions. Most recently, as a Supplementary Instructor in Biostatistics at the University of Nebraska Medical Center, I have worked directly with graduate students learning statistical methods for health research. Over time, I have learned (and am still learning) the small but important differences between teaching undergraduates new to quantitative reasoning, supporting graduate students in a technical methods course, and presenting research to colleagues at a conference. Teaching the same core ideas to undergraduates in Ghana, mathematics students in Ohio, and biostatistics graduate students in Nebraska has taught me one thing I now treat as central: the mathematics does not change, but the way students arrive at it does. In other words, I have experienced how one can be both a teacher and a student at the same time.

I believe statistics should give students more than technical skills. It should also teach them to think clearly, reason carefully, and communicate their findings honestly. Two convictions shape everything I do in the classroom. First, I want students to understand not only how a method works, but why it works and when it should be used. Students who learn statistics only as a sequence of steps forget those steps quickly. Students who understand the reasoning behind those steps become real statistical thinkers who can adapt when a new problem does not match the textbook. Second, I teach intuition before formalism. I want students to have a mental picture of what a method is doing before they meet its mathematical machinery. I run an interactive classroom where participation is expected and mistakes are treated as part of learning rather than as failures. I lean heavily on real public-health examples, because abstraction becomes memorable once it is attached to something students can care about. When I teach probability, for example, I use disease-screening scenarios. I ask students what a positive test result actually means for an individual patient. The mathematics of conditional probability suddenly matters, because the answer has real consequences. I also design courses for active engagement rather than passive listening. I use in-class exercises to check understanding in real time, structured group work so students learn from one another, and real datasets from health and environmental research rather than artificial examples.

Teaching has often been a rewarding experience for me, and one occasion has stayed with me as proof of what good support can do. I believe that almost any student can improve dramatically with the right help. A student in my class scored 24% on an early exam. Rather than letting them work it out alone, I met with them one-on-one outside of class. We worked through problem sets together. I gave specific feedback on exactly where their reasoning was breaking down: not just which answers were wrong, but which assumptions and steps were leading them astray. On the next exam, that same student scored 72%. What matters in that story is not only the grade. It is the shift from feeling lost to feeling capable. That kind of shift is what I work toward with every student. It is also why I treat office hours and individual feedback as central to teaching, not as an add-on to lectures.

As a first-generation college student and an international student of color in a quantitative field, I know how much representation and a sense of belonging shape whether a student speaks up or quietly disengages. I make equity a practical priority in concrete, classroom-level ways. I use anonymous polling tools such as TopHat so that students who are hesitant to speak publicly can still contribute their thinking without fear of judgment. I use collaborative review tools so that students with different learning styles have more than one way to engage with the material. I offer assessments in multiple formats, including problem sets, coding labs, short written reflections, and group projects. This way, students can show what they understand in the way that best fits how they think. I also design groups intentionally to bring together students from different backgrounds, because I have repeatedly seen how much they learn from one another when that diversity is present. I am especially attentive to students who are the first in their families to attend university. They are often less likely to ask for help, not because they need it less, but because they are not sure it is available to them. I make my availability explicit and repeated, because an open door only helps students who believe they are allowed to walk through it.

At the undergraduate level, I can teach a broad set of courses in biostatistics, probability, statistical methods, and statistical computing. At the graduate level, I can teach standard courses such as hierarchical and mixed models, categorical data analysis, and statistical computing in R, SAS, and Python. In addition, I would be eager to introduce two new advanced graduate-level courses.

- **Bayesian Statistics:** I will build on the experience I have gained as a researcher and instructor. I would begin with probability, Bayes' theorem, and the logic of updating beliefs with data. I would then build toward hierarchical models and MCMC. The course would keep intuition ahead of formalism. It would use real health and environmental datasets in R and NIMBLE, and culminate in a student-chosen final project.
- **Spatial Statistics and Environmental Biostatistics:** I intend to develop this course carefully. The goal is to expose students to the fundamental principles used to model health outcomes that vary across space and time. The course will draw on real disease-mapping and surveillance data, so that students learn to design, fit, and interpret spatial models in a way that is directly useful for public health research and policy. It will also engage students with three issues that will only grow in importance: reproducibility, the honest treatment of uncertainty, and the equity dimensions of environmental health.

I believe universities carry an important social responsibility to train and mentor students to keep "open minds" and to think rationally. My interactions with students across four institutions, together with my own positive experience of being mentored as a first-generation student, lead me to believe I am well prepared to do so. Teaching and mentoring give me the opportunity to understand, learn, and help shape capable young statisticians toward a larger cause: the betterment of both science and the communities that science is meant to serve. That cause is one I believe in deeply, and it is, in short, my central motivation for pursuing an academic position.